Neural Stiff-Chemistry Surrogates in DeepFlame: Implementation Pitfalls, Coupled-Solver Safeguards, and Target Semantics for H_2 and C_2H_4

Project draft generated from experiment record

April 24, 2026

Abstract

This draft manuscript summarizes the current state of an EFNO-oriented stiff-chemistry implementation and deployment program for H_2 and C_2H_4 inside DeepFlame. The strongest present story is not a final claim of solved CFD-coupled chemistry replacement, but an evidence chain showing: (1) EFNO-style stiff-chemistry surrogates are highly sensitive to implementation details, including a consequential transformed-state decode bug; (2) offline accuracy and rollout metrics do not directly predict coupled DeepFlame usefulness; (3) H_2 deployment requires thermodynamic diagnostics and guarded hybrid policies to remain solver-usable; and (4) for the more difficult C_2H_4 case, data construction and label semantics appear to dominate the remaining error budget. The manuscript is therefore organized as a reviewable partial-results paper with explicit negative results, deployment failures, and reproducible artifacts.

1 Introduction

Stiff chemistry surrogates are attractive for CFD because conventional chemistry integration remains expensive, but the deployment-facing evidence base for neural surrogates is still thin [2, 1, 4]. This project targets a faithful EFNO-style implementation study for H_2 and C_2H_4 inside DeepFlame, with emphasis on what survives coupled deployment rather than only what looks good offline.

2 Background and related work

The manuscript is grounded in the EFNO paper by Weng et al. [3] and a local-paper crosswalk covering data generation, augmentation, and coupled-validation issues from related combustion-chemistry learning papers [2, 5, 1, 4]. Across these papers, a recurring theme is that model architecture alone is not decisive: data semantics, postprocessing contracts, augmentation strategy, and a posteriori validation all materially affect conclusions.

3 Reproducible workflow and software stack

The implementation workflow combines PDF extraction and structured notes, DFODE-kit extensions for EFNO-style training and evaluation, and DeepFlame export bridges for MLP and FNO-like surrogates. The software path is deliberately aligned with the emerging DFODE-kit/DeepFlame ecosystem [1, 2]: data are prepared from canonical or case-aligned thermochemical states, models

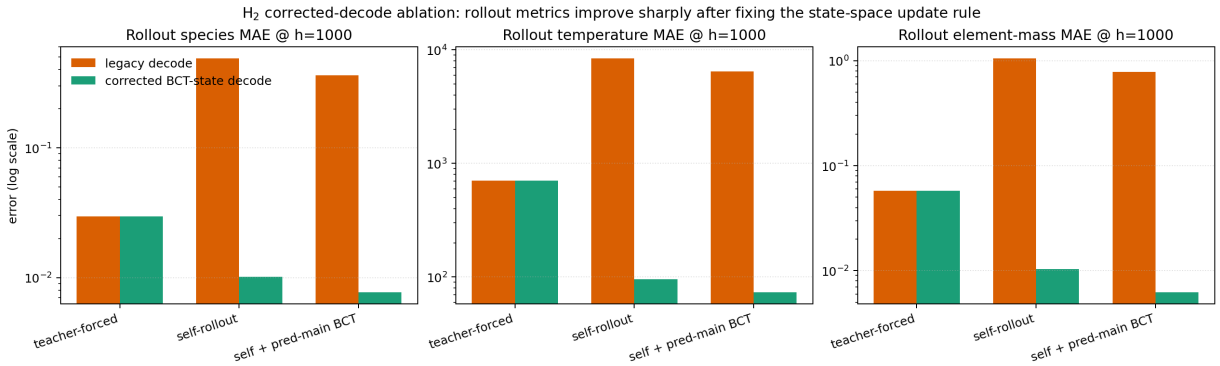

Figure 1: H_2 legacy-vs-corrected decode ablation on rollout metrics at horizon 1000. Fixing the transformed-state update rule changes the empirical conclusion most strongly for the self-rollout branches, supporting the claim that a large part of the earlier instability was implementation-induced rather than intrinsic to self-feeding.

are trained and evaluated offline with explicit rollout and physical-consistency metrics, and promising checkpoints are exported into DeepFlame for coupled testing. All substantive experiments are documented in the project docs site together with scripts and artifact paths.

4 Offline H_2 reproduction and implementation diagnosis

The H_2 program established seeded offline baselines, rollout-aware EFNO-style training, and a key implementation diagnosis: a transformed-state species decode bug materially distorted rollout behavior. After correcting the update rule to act in the proper transformed state space, self-rollout behavior changed qualitatively and the corrected branch became a credible offline reference for deployment-facing work. Figure 1 shows that the correction was not a small numerical cleanup: it sharply reduced long-horizon rollout error, especially for the self-rollout branches.

5 DeepFlame-coupled H_2: from failure to guarded deployment

The H_2 coupled study showed that export correctness and offline metrics are not enough. Coupled failures appeared first as HP reconstruction breakdowns, which led to hybrid fallback policies, frozen-temperature gating, and a logistic risk guard. Figure 2 shows how threshold and guard choices reshape the coupled operating regime over time. Table 1 summarizes representative operating points for the corrected branch. In the present evidence base, 650 K gives the strongest long-horizon single-threshold retention, while the logistic risk guard at threshold 0.5 provides a more safety-oriented mode with lower HP-failure exposure at some cost in learned participation.

The H_2 evidence supports a reviewable claim that deployment-facing chemistry surrogates require contract alignment and explicit thermodynamic safeguards, not just better offline loss values.

6 C_2H_4 baseline establishment and runtime bridge diagnosis

For C_2H_4, the program first established a trustworthy stock DeepFlame baseline, diagnosed reduced-rank GPU pitfalls, and repaired the FNO runtime bridge by batching inference. These steps were necessary before data-quality conclusions could be trusted: without them, runtime pathologies

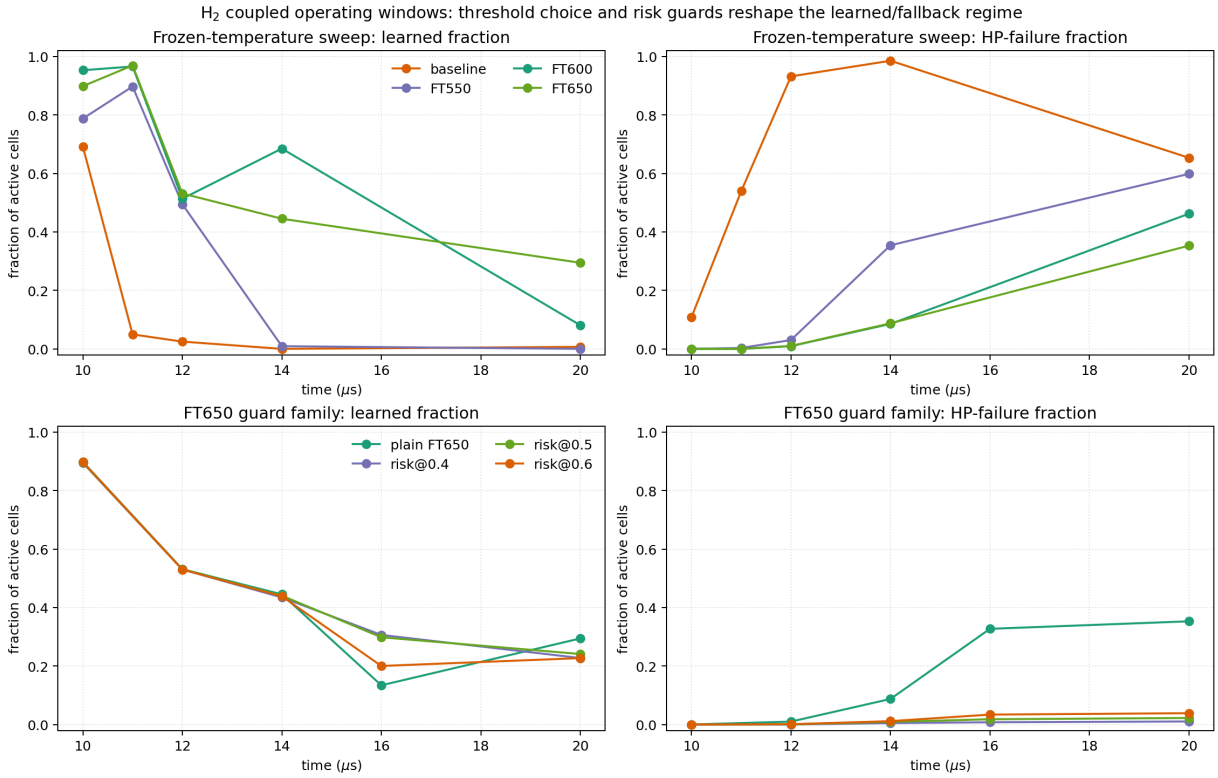

Figure 2: H_2 coupled operating windows from the corrected branch. Top row: frozen-temperature sweep showing learned-cell retention and HP-failure fractions over time. Bottom row: FT650 risk-guard family. The figure makes two manuscript-level points explicit: raising the activation threshold delays collapse materially, and the logistic risk guard trades some learned participation for much lower HP-failure incidence.

could masquerade as model deficiencies. The emphasis on a posteriori validation follows the practice advocated in recent combustion-surrogate studies, which consistently show that apparently strong offline models can still fail once coupled back into CFD [2, 5, 4].

7 C_2H_4 data semantics: what the current best model still gets wrong

The strongest current C_2H_4 data result is a calibrated mixed dataset that combines the solver-aligned $dp100$ case-pair backbone with a moderate canonical augmentation component. This mixed model survives to 5×10^{-6} and is substantially better than earlier baselines, but the best-model-vs-stock anatomy shows a structured residual failure: cooler active cells are over-driven, the hot bulk exhibits widespread source-term sign errors, and several late intermediates are still heavily suppressed. The mixed-data direction itself is motivated by nearby literature showing the value of canonical flame manifolds, interpolation-based densification, and physics-aware perturbation for building thermochemical training sets that cover reactive regions more effectively [2, 5, 1].

The next diagnostic step showed that the target-semantics hypothesis is real in the data itself: one-step chemistry-proxy relabeling of case-pair current states produces materially different targets, especially on key intermediates. Table 3 summarizes the most distorted active-region species in the

Table 1: Selected H_2 coupled-deployment operating points for the corrected branch in DeepFlame.

Mode	Learned frac. at 2×10^{-5}	Cum. fallback frac. at 2×10^{-5}	First time > 95% fallback
FT550	0.000	0.516	1.4e-05
FT600	0.081	0.373	–
FT650	0.294	0.361	–
FT650+risk@0.5	0.241	0.381	–

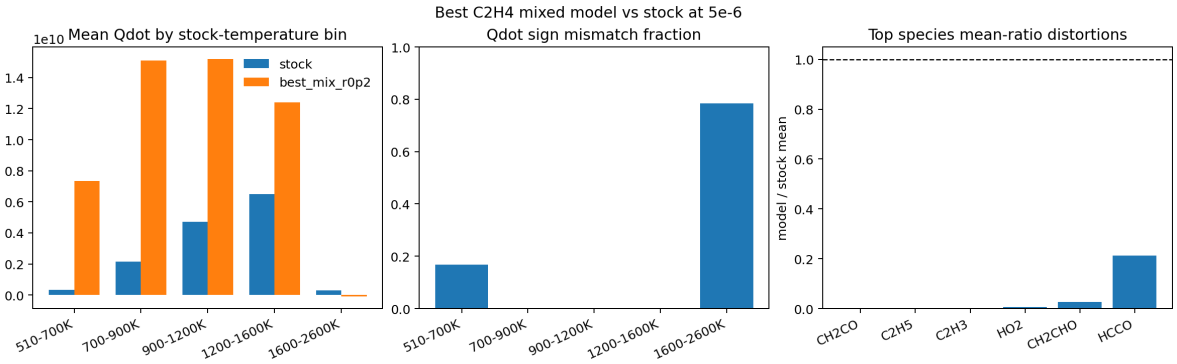

Figure 3: Matched-mesh comparison between the current best mixed C_2H_4 model and the stock DeepFlame baseline at 5×10^{-6}. The figure highlights temperature-binned source-term mismatch and the strongest mean-ratio species distortions.

current best mixed model, while Table 4 shows that the original CFD next-state labels differ materially from chemistry-proxy relabels on a sampled subset. However, deployment tests also showed that semantics-only fixes are not enough in isolation. A tiny pure chemistry-proxy model fails catastrophically by 4×10^{-7}, while naive partial replacement of 10% of labels preserves runtime survival but badly degrades solver-facing quality.

Table 2 makes the current C_2H_4 claim boundary explicit. The best mixed model is reviewably better than prior baselines, but naive partial chemistry-proxy relabeling is not yet a deployment improvement. Tables 3 and 4 sharpen the interpretation: the remaining gap is tied both to suppressed intermediate chemistry in deployment and to a measurable mismatch between full-CFD and chemistry-only target semantics. This supports the more specific forward hypothesis that semantics must be improved without losing the solver-relevant manifold.

8 Discussion

The present evidence supports several broader lessons. First, transformed-state implementation details can dominate empirical conclusions, which reinforces the need to treat reproduction as implementation diagnosis rather than a narrow hyperparameter chase [3]. Second, offline and deployment-facing evaluations can disagree substantially because of postprocessing and thermodynamic contract issues, in line with the broader combustion-surrogate literature's emphasis on a posteriori validation [2, 4]. Third, for the more difficult C_2H_4 case, the dominant remaining bottleneck appears to be target construction rather than another round of small architectural or mixing sweeps.

Table 2: C_2H_4 deployment-facing summary at 5×10^{-6} for the stock baseline, the current best mixed model, and a naive partial chemistry-proxy relabeling ablation.

Model	Mean $\dot{Q}$	$\dot{Q}$/stock	$p_{\max}$ (Pa)	$T_{\min}$ (K)	mean $\lvert\Delta T\rvert$ (K)
Stock DeepFlame	1.62e+07	1.000	1.02e+05	499.249	0.828
Best mixed ($dp100+$ $canon@0.2$)	2.51e+07	1.551	1.03e+05	499.159	0.927
Partial chem-proxy (10%)	-9.88e+08	-60.952	1.03e+05	492.524	9.153

Table 3: Largest active-region species distortions for the current best mixed C_2H_4 model relative to stock at 5×10^{-6}.

Species	Stock mean	Model mean	Model/stock
CH2CO	2.12e-04	2.95e-17	1.39e-13
C2H5	1.50e-05	2.22e-16	1.47e-11
C2H3	5.81e-05	3.45e-10	5.95e-06
HO2	5.46e-05	3.64e-07	0.007
CH2CHO	6.63e-06	1.77e-07	0.027
HCCO	6.01e-06	1.28e-06	0.212

9 Conclusion

The current project state is strong enough for a reviewable partial-results paper. For H_2, corrected implementation plus guards yields a credible deployment story. For C_2H_4, the strongest current result is a calibrated mixed-data baseline together with a clear diagnosis that better target semantics are needed, but must be introduced without abandoning the solver-relevant state manifold. The central take-home message is that offline surrogate success is insufficient without deployment-aware validation and explicit thermodynamic diagnostics.

Immediate remaining tasks before submission

1. Add citations from the extracted EFNO notes and local related papers.
2. Package at least one H_2 offline corrected-decode figure and one H_2 coupled-timeline figure.
3. Expand the manuscript body from a structured skeleton into a fully referenced narrative.
4. Replace naive semantics-only C_2H_4 relabeling with a more selective semantics-improvement experiment, or scope that step explicitly as future work.

References

[1] Han Li, Ke Xiao, Yangchen Xu, Haoze Zhang, Zhenyi Chen, Runze Mao, and Zhi X. Chen. DFODE-Kit: Deep learning package for solving flame chemical kinetics with high-dimensional stiff ordinary differential equations. *Computer Physics Communications*, 321:110013, 2026.

Table 4: Largest mean-ratio shifts between original CFD next-state labels and chemistry-proxy relabels on a 5k-row C_2H_4 subset.

Species	Original next mean	Chemistry-proxy next mean	Chem/original
HCCO	3.82e-07	1.34e-05	35.146
C2H5	5.13e-06	2.12e-07	0.041
CH2CHO	2.54e-06	1.08e-05	4.269
CH2OH	1.38e-06	5.13e-06	3.713
C2H3	3.90e-05	1.03e-04	2.651
CH2CO	6.24e-05	1.64e-04	2.621

[2] Han Li, Ruixin Yang, Yangchen Xu, Min Zhang, Runze Mao, and Zhi X. Chen. Comprehensive deep learning for combustion chemistry integration: Multi-fuel generalization and a posteriori validation in reacting flow. *Physics of Fluids*, 37:015162, 2025.

[3] Yuting Weng, Han Li, Hao Zhang, Zhi X. Chen, and Dezhi Zhou. Extended Fourier neural operators to learn stiff chemical kinetics under unseen conditions. *Combustion and Flame*, 272:113847, 2025.

[4] Sipei Wu, Wenkai Liang, and Kai Hong Luo. Deep learning based combustion chemistry acceleration method for widely applicable NH_3/H_2 turbulent combustion simulations. *Combustion and Flame*, 278:114218, 2025.

[5] Ke Xiao, Yangchen Xu, Han Li, and Zhi X. Chen. Enhancing deep learning of ammonia/natural gas combustion kinetics via physics-aware data augmentation and scale separation. *Fuel*, 419:138904, 2026.